Finite-size decay-mode hybridization near a mean-field fold in an open Rydberg ensemble

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How does finite-size decay-mode mixing near a mean-field fold appear in physical relaxation? We study a uniformly interacting Rydberg ensemble with independent local decay, retaining the complete permutation-invariant operator space. Two real decay eigenvalues exhibit an avoided crossing on the monostable side of the fold, with an approximately exponential decrease of their separation over $N = 24$–56. Physical product-state perturbations identify an exchange of high-branch relaxation residues between modes whose common decay rate remains finite. A biorthogonal reduction identifies the invariant coupling product, while exact coherence elimination recovers both eigenvalues as self-consistent roots of one effective branch. Reconstructing observable residues additionally requires the initial-coherence source and spectral-derivative normalization. Fixed-preparation controls at $N = 24$–48 distinguish this spectral exchange from the complete density transient. Over $0 \le \gamma t \le 12$, the sampled signals remain smooth across the narrow mixing region, despite substantial changes of individual residues. The coherence-to-population peak is smaller than its independent-atom counterpart, although mean field fails to reproduce the complete transient. These results quantify the observable limits of the finite-size spectral structure without deriving an asymptotic switching action or fold correspondence.

I. INTRODUCTION

Driven Rydberg ensembles connect nonlinear mean-field bistability with finite quantum dynamics [1–4]. A stable mean-field branch terminates at a saddle-node fold, whereas a finite quantum system evolves under a linear, trace-preserving Liouvillian. Relating these descriptions requires distinguishing stationary mean-field roots, eigenvalues of their stability Jacobian, and eigenvalues of the many-body generator.

This distinction matters for exceptional-point (EP) interpretations of nonlinear switching and stationary-root singularities [4–6]. A fold of stationary solutions need not be a defective eigenvalue of the mean-field Jacobian. The stationary sector of a finite-dimensional quantum dynamical semigroup cannot contain a nontrivial Jordan block, which would produce unbounded polynomial growth [7]. In the spectral-separation framework, metastability involves low-lying decay modes separated from faster relaxation [8, 9]. More generally, large eigenmode-expansion coefficients can prolong transients, so decay eigenvalues alone do not determine relaxation times [10, 11].

For uniformly interacting spins with local decay, collective switching and steady-state correlations have been studied [1, 2], and an exact finite-N steady state is available [12]. Exponentially slow switching is central to first-order dissipative transitions [13]; quantum-instanton methods predict switching rates for collective spin systems, including local dissipation [14, 15]. A related conjecture connects a symmetry-selected steady-state purification potential to the ordinary Liouvillian gap [16]. These results concern relaxation toward the stationary sector, rather than the separation of two modes with a finite common decay rate.

Finite-size avoided crossings from mode mixing [17] and few-mode descriptions of hysteresis [18] also have precedents. Cabot, Giorgi, and Zambrini connected a Liouvillian EP to a mean-field bifurcation between phase diffusion and quantum activation in a driven oscillator [19]. Collective-spin models exhibit spectral doublets approaching finite nonzero decay rates [20], with preparation-dependent visibility and cancelling spectral residues [21]. These precedents leave a model-specific question for the locally damped Rydberg ensemble: how does a real-mode avoided crossing redistribute response, and does that redistribution produce a sharp density transient?

We address this question at $\Omega = 2\gamma$ and $V = 10\gamma$ in the complete permutation-invariant operator space. We first identify mode character using physical preparations and signed observable residues. Biorthogonal projection and coherence elimination then connect the splitting to its effective roots and response weights. Finally, fixed-preparation scans and independent-atom and mean-field controls test the implications for complete transients. This separates the finite-size spectral structure from claims about observable enhancement or asymptotic switching.

II. MODEL, MEAN-FIELD FOLD, AND FINITE-SIZE SPECTRUM

We consider N two-level atoms with excited-state number $\hat{N}_e = \sum_i \hat{n}_i$, collective spin $\hat{J}_\alpha = \frac{1}{2}\sum_i \hat{\sigma}_i^\alpha$, and independent decay:

$$\dot{\rho} = \mathcal{L}_N \rho = -i[\hat{H}, \rho] + \gamma \sum_i \mathcal{D}[\hat{\sigma}_i^-]\rho, \tag{1}$$

$$\hat{H} = -\Delta \hat{N}_e + \Omega \hat{J}_x + \frac{V}{2N}\hat{N}_e(\hat{N}_e - 1), \tag{2}$$

where $\mathcal{D}[c]\rho = c\rho c^\dagger - \{c^\dagger c, \rho\}/2$. This is the uniformly interacting, locally damped model of Refs. [1, 2], within

the exactly solvable steady-state family of Ref. [12]. The Supplemental Material (SM) gives the finite-N parameter mappings, including the detuning shift associated with $\hat{N}_e(\hat{N}_e-1)$. We set $\gamma=1$, so frequencies and eigenvalues below are measured in units of γ, and time in γ^{-1}.

Permutation invariance reduces the operator space to dimension $\binom{N+3}{3}$ [22]. We retain every Dicke j block and the transitions between blocks caused by local decay. This differs from the maximum-spin effective-jump approximation used in recent Rydberg switching calculations [23]. Throughout, "full generator" denotes this complete permutation-invariant operator space, which contains the product states and observables considered here. It does not enumerate modes in other permutation-symmetry sectors.

In the large-N factorized limit [1, 2], write the single-atom coherence as $\rho_{eg}=u+iv$ and the excited population as n. The mean-field equations are

$$\begin{aligned}\dot{n}&=-n-\Omega v,\\ \dot{u}&=-(\Delta-Vn)v-u/2,\\ \dot{v}&=(\Delta-Vn)u-\Omega(1-2n)/2-v/2.\end{aligned}\tag{3}$$

Their stationary populations satisfy

$$F(n)=n[(\Delta-Vn)^2+1/4+\Omega^2/2]-\Omega^2/4=0.\tag{4}$$

At $\Omega=2$, $V=10$, the left fold is at

$$(\Delta_f,n_f)=(3.5996368905,\,0.1569270403).\tag{5}$$

Using $F=\partial_n F=0$, the characteristic polynomial of the Bloch Jacobian at a fold reduces to

$$\chi(\lambda)=\lambda[(\lambda+1)^2+\Omega^2/2].\tag{6}$$

Thus the zero eigenvalue is simple, including when the fold reaches the cusp. Li, Sun, and Yi [6] obtained square-root and cube-root singularities of stationary excitation densities near fold and cusp points. Those exponents describe roots of the nonlinear stationary equation; Eq. (6) concerns the distinct linear stability problem. In particular, a triple stationary root does not imply an EP3 of this Jacobian. The coalescing fixed points and the surviving high-population branch must also be distinguished; Eq. (6) describes the former.

We locate a local minimum of the separation between two real decay eigenvalues and denote it by $\Delta_*(N)$. Define

$$g_N=\frac{\lambda_+(\Delta_*)-\lambda_-(\Delta_*)}{2}>0.\tag{7}$$

The positions move from $\Delta_*(24)=3.13740295$ to $\Delta_*(56)=3.30652148$, and the half-gap decreases from 0.0292933152 to 0.000330979777 [Fig. 1]. A least-squares slope of $\ln g_N$ over $N=24,28,\ldots,56$ gives $\sigma_{24:56}=0.14050$. Restricting to $N=40$–56 gives 0.14305, while successive four-atom slopes increase from 0.13218 to 0.14383. We therefore use "approximately exponential" only for the observed size window. The data do not determine a limiting exponent or prove $\Delta_*(N)\to\Delta_f$.

The small splitting must not be confused with a slow decay rate. At $N=32$ and $\Delta_*=3.205747775$, the real pair is

$$\lambda_-=-0.765098904,\qquad \lambda_+=-0.745092273,\tag{8}$$

whereas an oscillatory pair occurs at

$$\lambda_{\rm osc}=-0.725036287\pm 2.864794973i.\tag{9}$$

The oscillatory pair decays more slowly than both real modes. Thus g_N is an excited-mode separation, distinct from the ordinary Liouvillian gap $\Gamma_{\rm diss}=-\max_{\lambda\neq0}\mathrm{Re}\,\lambda$. It does not measure the slowest relaxation toward the stationary sector.

III. PHYSICAL RESPONSE AND MODE CHARACTER

A Hermitian right eigenoperator at a nonzero decay eigenvalue is traceless; its signed excitation-number profile is not a probability distribution. We test mode character through observable residues, whose dependence on both preparation and detection is essential in nonnormal dynamics [10, 21].

Let R_i and W_i be right and transpose-left eigenvectors of the stored generator, normalized by $W_i^TR_j=\delta_{ij}$. The transpose denotes the algebraic dual of the chosen vectorization. For the excitation-density observable $o^T\rho=\langle\hat{N}_e\rangle/N$, an initial difference $\delta\rho$ gives the exact contribution of a simple mode,

$$\delta n_i(t)=a_ie^{\lambda_it},\qquad a_i=(o^TR_i)(W_i^T\delta\rho).\tag{10}$$

Each residue is invariant under reciprocal rescaling of its left and right eigenvectors. These signed weights are not probabilities of physical states or metastable phases.

We use two physical preparations, whose construction is specified in the SM. The first subtracts positive product states displaced by $\pm\eta$, with $\eta=0.005$, along the real relaxation direction of the high mean-field fixed point. This high-branch preparation is reconstructed at each detuning to follow local mode character. The second subtracts the exact stationary state from a fixed product state prepared at the left-fold Bloch vector. It probes low-population response without assuming a finite-size metastable state outside the bistable interval.

For $N=32$, at $\delta=\Delta-\Delta_*=-0.06$, the high-branch perturbation has real-mode residues

$$(a_-,a_+)=(8.94\times10^{-6},\,1.1694\times10^{-3}).\tag{11}$$

At $\delta=+0.06$, they become

$$(a_-,a_+)=(1.1082\times10^{-3},\,-2.91\times10^{-6}).\tag{12}$$

The dominant high-branch response transfers from λ_+ to λ_- [Fig. 2(a)]. A separate $N=40$ calculation gives

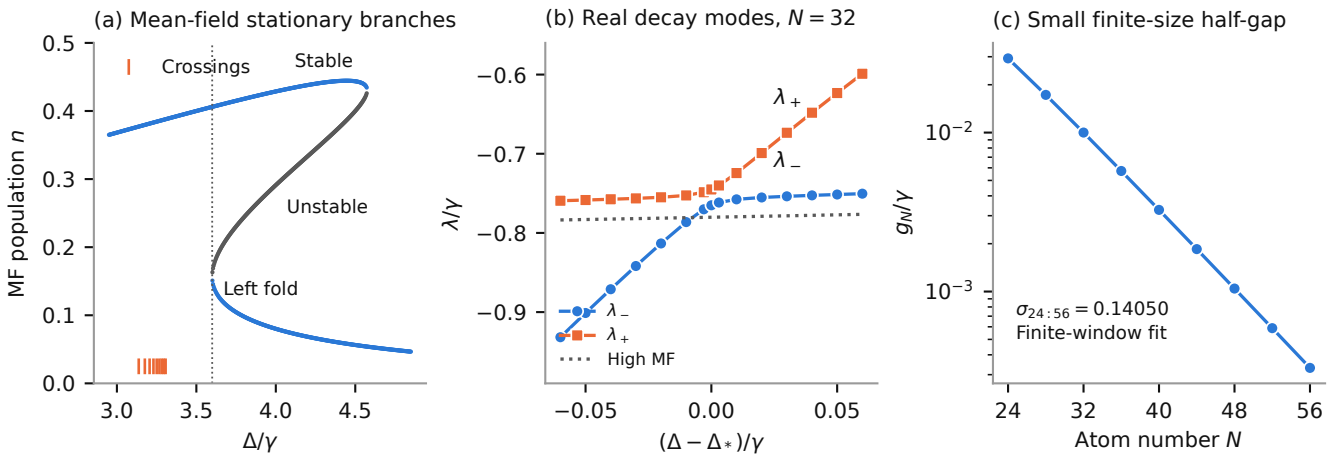


FIG. 1. Mean-field geometry and the finite-size real-mode crossing at $\Omega = 2$, $V = 10$. (a) Stationary populations: stable branches in blue, unstable branch in gray. Orange ticks mark the real-axis crossing positions for $N = 24, 28, \ldots, 56$; all lie on the monostable side of the left fold. (b) The two target real eigenvalues for $N = 32$, ordered as $\lambda_- < \lambda_+$, versus $\delta = \Delta - \Delta_*$. The dotted curve is the real relaxation eigenvalue of the surviving high mean-field branch. The modes exchange high-branch response, but the finite-size rates need not equal the mean-field rate. (c) Half-gap at its local minimum. The quoted exponent is a finite-window fit, not an established large-N limit.

the same exchange, with the nondominant residue below 7.3×10^{-7} at either endpoint. Halving η changes the normalized residue vector by less than 7.3×10^{-6} in relative Euclidean norm.

The fold-state preparation excites both modes, with large opposite residues near the crossing [Fig. 2(b)]. At $\delta = 0$, they are approximately $(-1.666, 1.423)$ for $N = 32$ and $(-5.475, 5.217)$ for $N = 40$. Opposite residues can cancel in a bounded physical signal, as analyzed for collective-spin exceptional spectral phases [21]. Our values establish neither a $1/g_N$ residue asymptote nor a real-axis Jordan limit; the SM states the additional limiting conditions.

The second diagnostic is the signed excitation-number profile $P_i(k) = \mathrm{Tr}(\Pi_k R_i)$. We measure its absolute-profile fraction in the region $k/N < 0.22$, using the same fixed diagnostic cutoff on both sides. For $N = 32$, the fractions change from $(0.209, 0.0305)$ at $\delta = -0.06$ to $(0.0314, 0.237)$ at $\delta = +0.06$ [Fig. 2(c)]. Thus the branch with greater low-density support exchanges places with the high-branch response. Neither profile is purely localized in that region. We describe the second branch as having enhanced low-density character; a rigorous identification with a universal ghost-escape mode would require an independently controlled escape problem and is not assumed here.

IV. BIORTHOGONAL TWO-MODE REDUCTION AND COMPLEX-DETUNING EPS

A. Local reduction at the gap minimum

Write $\mathcal{L}_N(\Delta) = L_0 + \Delta L_1$, with $L_1\rho = i[\hat{N}_e, \rho]$. At Δ_*, choose bases R, W of the exact two-dimensional right and left invariant subspaces with $W^T R = I$. Diagonalizing $W^T L_1 R$ defines a local diabatic basis with distinct real slopes s_a, s_b. Holding that basis fixed gives

$$H_P(\delta) = \begin{pmatrix} e_a + s_a\delta & V_{ab} \\ V_{ba} & e_b + s_b\delta \end{pmatrix}, \qquad \delta = \Delta - \Delta_*. \quad (13)$$

Here a denotes the smaller-absolute-slope branch. At $\delta = 0$ the projection is exact. Coupling to the complementary invariant subspace produces a spectral self-energy starting at order δ^2 (SM); its small absolute size does not alone establish its smallness relative to an exponentially small coupling.

The usual two-by-two discriminant [24] gives the squared separation and its derivative at the projection point:

$$d^2 = (e_a - e_b)^2 + 4V_{ab}V_{ba}, \qquad \partial_\Delta d^2 = 2(e_a - e_b)(s_a - s_b). \quad (14)$$

At a nondegenerate minimum of a positive real gap, distinct slopes imply $e_a = e_b$ and hence

$$\boxed{g_N^2 = V_{ab}V_{ba}.} \quad (15)$$

This identity uses the exact finite-size spectral subspace. It organizes the gap but is not an independent prediction of its exponent.

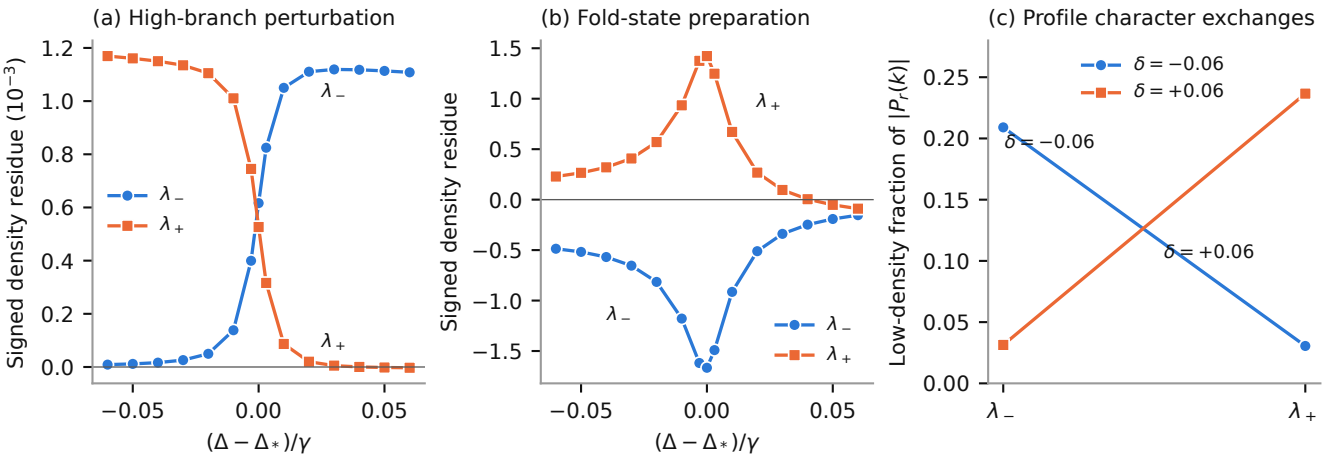


FIG. 2. Physical residues and exchange of mode character, $N = 32$. (a) Residues for product-state differences about the high mean-field fixed point, reconstructed at each detuning. (b) Residues for a fixed fold product state relative to the current stationary state. Large opposite residues cancel in the physical signal. (c) Fraction of the absolute excitation-number profile below $k/N = 0.22$ at the two endpoints. This normalization-independent ratio characterizes a signed eigenoperator, not a population probability of a physical state.

Under $R_i \mapsto c_i R_i$, $W_i \mapsto c_i^{-1} W_i$,

$$V_{ab} \mapsto (c_b/c_a)V_{ab}, \qquad V_{ba} \mapsto (c_a/c_b)V_{ba}. \tag{16}$$

Individual coupling exponents can therefore depend on normalization if c_i depend on N. We fix a population-peak convention, $\max_k |P_i(k)| = 1$, with the largest-magnitude entry positive. Over $N = 24$–56, the descriptive exponents are

$$\begin{aligned} \alpha_{ab} &= 0.009638, \qquad \alpha_{ba} = 0.271365, \\ (\alpha_{ab} + \alpha_{ba})/2 &= 0.140501. \end{aligned} \tag{17}$$

Normalizing the right operators by their physical trace norm changes the first two values to 0.003563 and 0.277440, while preserving their sum. A balanced convention makes both equal to σ by construction [Fig. 3]. Thus a directional exponent cannot be assigned an invariant activation interpretation from this decomposition alone.

The weak matrix element contains cancellations involving coherences. For $N = 40$, summing its output-diagonal and output-off-diagonal contributions separately gives

$$V_{ba} = 0.9659542705 - 0.9657905699 = 1.63700625 \times 10^{-4}. \tag{18}$$

This is not a representation-independent separation into two physical processes. Indeed, $W_b^T(L - cI)R_a = W_b^T L R_a$ for any scalar c, and choosing $c = e_a$ moves the cancellation into the small residual $(L - e_a)R_a$. The robust conclusion is that a right population tail by itself does not determine the full biorthogonal matrix element. A boundary-flux identity and its dependence on the chosen cut are given in the SM.

B. Analytic continuation in detuning

We separately continue the full generator as $L_0 + \zeta L_1$ for complex detuning ζ. This continuation uses $-i[H(\zeta), \rho]$; it is not $-i[H\rho - \rho H^\dagger]$. The continued evolution preserves trace but generally not Hermiticity. It is therefore not a completely positive and trace-preserving (CPTP) physical map.

A local two-subspace Newton calculation gives the upper-half-plane EP positions

$$\begin{aligned} \zeta_{\mathrm{EP}}(24) &= 3.136806063 + 0.026370766i, \\ \zeta_{\mathrm{EP}}(32) &= 3.205686613 + 0.007464558i, \\ \zeta_{\mathrm{EP}}(40) &= 3.250218904 + 0.002129005i. \end{aligned} \tag{19}$$

At each point, the projected traceless matrix M has nonzero norm, while $\|M^2\|_F / \|M\|_F^2 < 3.3 \times 10^{-15}$, consistent with a rank-one nilpotent. For $N = 32$, analytic continuation around a circle of radius 0.00223937 exchanges the two full-generator eigenvalues after one circuit and restores them after two. Changing the eigensolver seed and shift and doubling the sampling density preserves the result (SM).

The local linear model predicts $|\operatorname{Im}\zeta_{\mathrm{EP}}| \simeq 2g_N/|s_a - s_b|$. The relative differences in the imaginary parts are 0.456%, 0.0792%, and 0.0115% for $N = 24, 32, 40$, respectively. This verifies nearby complex partners for the studied crossings; we do not claim that an arbitrary avoided crossing necessarily has a nearby EP. Nor does an EP in complex detuning constitute a physical degeneracy at real parameters.

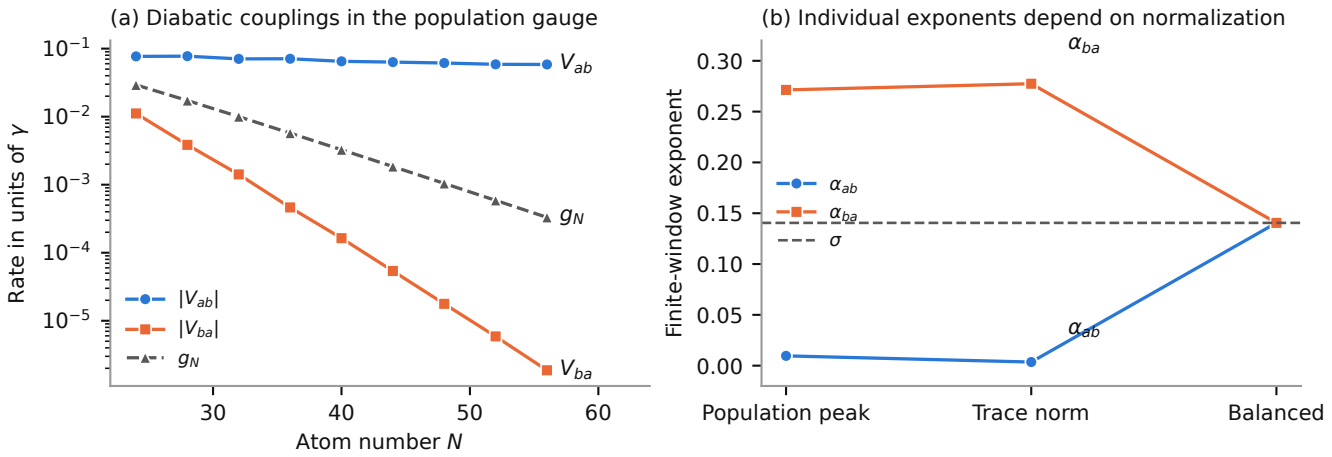


FIG. 3. Biorthogonal gap decomposition. (a) Couplings in the population-peak convention, together with the invariant half-gap. (b) Finite-window exponents under three normalizations of the same two-dimensional subspace. The physical trace-norm convention changes the individual exponents; balancing the couplings makes each equal to σ by construction. The invariant statement is their product and the sum of their logarithmic slopes.

V. EXACT ELIMINATION OF COHERENCE SECTORS

A. Spectral-parameter-dependent effective operator

We use projection methods that retain the spectral parameter [25, 26], evaluated by matrix continued fractions as in spin master equations [27]. Group the Dicke operator basis by signed coherence order $q = m - m'$. The diagonal Hamiltonian and local decay preserve q, whereas the drive connects q only to $q \pm 1$. This gives a block-tridiagonal generator while retaining every Dicke j block.

Let $A_q = L_{q,q}$, $U_q = L_{q,q+1}$, and $D_q = L_{q+1,q}$. Starting at the highest positive order $q_{\mathrm{top}} = N$, define

$$\Sigma_{q_{\mathrm{top}}}(z) = 0,$$
$$\Sigma_{q-1}(z) = U_{q-1}[zI - A_q - \Sigma_q(z)]^{-1} D_{q-1}. \tag{20}$$

Here z is an eigenvalue variable, distinct from the complex detuning ζ above. For real physical parameters and real z, Hermitian conjugation relates the positive and negative coherence chains. Their sum yields

$$K_N(z) = A_0 + 2\,\mathrm{Re}\,\Sigma_0(z), \qquad K_N(z) r_0 = z r_0. \tag{21}$$

The $q = 0$ sector still contains the populations of every Dicke block; it is not the excitation-number marginal alone, and $K_N(z)$ need not be a classical Markov generator.

With P projecting onto $q = 0$ and $Q = I - P$, the Schur identity reads

$$\det(zI - L) = \det(zI - L_{QQ})\det[zI - K_N(z)]. \tag{22}$$

It applies where $z - L_{QQ}$ is invertible; the recursion also requires invertible intermediate denominators. Purely coherent modes with zero $q = 0$ projection and points at elimination poles require separate treatment. For complex z, the analytic continuation involves $\Sigma_+(z) + \overline{\Sigma_+(\bar{z})}$, not $2\,\mathrm{Re}\,\Sigma_+(z)$. Our real-axis recursion is not used to compute Eq. (19).

The spectral dependence has a time-domain counterpart in the projection-operator memory formalism [28, 29]. Direct elimination of x_Q from our linear equations gives

$$\dot{x}_0(t) = L_{00} x_0(t) + L_{0Q} e^{t L_{QQ}} x_Q(0) + \int_0^t L_{0Q} e^{(t-s) L_{QQ}} L_{Q0} x_0(s)\, ds. \tag{23}$$

Thus the self-energy represents memory carried by the eliminated coherence sectors. Replacing it by a constant is an additional approximation.

B. Two roots of one effective branch

In the local real spectral window, select a simple real eigenvalue branch $\mu_N(z)$ of $K_N(z)$ and define $f_N(z) = \mu_N(z) - z$. We find a negative local minimum,

$$f_N(z_c) = -\epsilon_N < 0, \qquad f_N'(z_c) = 0,$$
$$c_N = f_N''(z_c) > 0. \tag{24}$$

Its two nearby roots give the two decay eigenvalues. Expanding around the minimum gives

$$f_N(z) = -\epsilon_N + \frac{c_N}{2}(z - z_c)^2 + \cdots, \qquad g_{N,\mathrm{quad}} = \sqrt{\frac{2\epsilon_N}{c_N}}. \tag{25}$$

The derivatives are calculated analytically through the matrix recursion and the non-Hermitian effective eigenproblem (SM).

TABLE I. Local real-axis coherence reduction. g_N is the independently computed full-generator half-gap; the last column is $g_{N,\mathrm{quad}}/g_N - 1$. All entries use $\gamma = 1$.

N	ϵ_N	c_N	g_N	Relative error
24	9.33404×10^{-3}	22.28695	0.02929332	-1.2002×10^{-2}
32	1.09619×10^{-3}	21.96534	0.01000332	-1.2773×10^{-3}
40	1.28422×10^{-4}	24.11955	0.00326360	-1.0842×10^{-4}
48	1.51689×10^{-5}	27.82335	0.00104421	-6.5891×10^{-6}

For $N = 24, 32, 40, 48$, direct roots of $f_N = 0$ reproduce the independent full-generator half-gaps to relative differences below 1.3×10^{-9}. Back-substituting all eight roots into the full generator gives $\|LR - zR\|_2/\|R\|_2 < 1.2 \times 10^{-13}$. Table I and Fig. 4 show the separate, larger error of the quadratic approximation. The minimum location is not generally the midpoint of the two roots: at $N = 24$, these are $z_c = -0.745502759$ and $(z_- + z_+)/2 = -0.750910792$.

The two roots do not correspond to two nearby eigenvalues of a frozen $K_N(z_c)$. For $N = 48$, its nearest eigenvalues include -0.75910208, -0.21846755, and zero, whereas the full generator has the two roots -0.76013902 and -0.75805059. The root pair is retained by self-consistency in z. This finding concerns this specific elimination; it does not rule out a different, correctly matched finite-dimensional effective model.

A four-point fit of $\ln \epsilon_N$ gives $\alpha_\epsilon = 0.267635$, while the same convention applied to the curvature gives $\alpha_c = -0.009490$. Consequently the quadratic gap has fitted exponent $(\alpha_\epsilon - \alpha_c)/2 = 0.138562$, compared with 0.139030 for the exact gap over the same four sizes. These are different fit windows from $\sigma_{24:56}$ above. No argument yet fixes the large-N scaling of ϵ_N or proves that c_N contributes only an algebraic prefactor. Equation (25) is therefore a tested finite-size description, not an action-based exponent prediction.

C. Connection to biorthogonal self-orthogonality

The effective spectral slope has a precise relation to the full-mode norm. Write $G = (zI - L_{QQ})^{-1}$, $B = L_{0Q}$, and $C = L_{Q0}$, so that $K = L_{00} + BGC$ and $K_z = -BG^2C$. For a simple effective eigenbranch with $Kr = \mu r$, $l^T K = \mu l^T$, and $l^T r = 1$, define the back-substituted vectors

$$R = \begin{pmatrix} r \\ GCr \end{pmatrix}, \qquad W^T = (l^T,\, l^T BG). \tag{26}$$

Then

$$\boxed{W^T R = l^T (I - K_z) r = 1 - \mu_z = -f_z.} \tag{27}$$

At a root, this is the standard derivative pairing for the nonlinear eigenvalue problem $T(z) = zI - K(z)$ [26, 30]. At the negative minimum, $f_z = 0$ but $f \neq 0$: the back-substituted vectors are not eigenmodes, so this is not an EP. Under regular elimination and a simple effective branch, $f = f_z = 0$ with $f_{zz} \neq 0$ would instead give an EP2 (SM). This criterion does not determine the observed depth ϵ_N.

D. Initial coherences and observable residues

The same elimination must retain the initial-coherence source in Eq. (23). For an initial operator $x = (x_0, x_Q)^T$, its reduced Laplace transform obeys

$$[zI - K(z)]\widetilde{x}_0(z) = b(z), \qquad b(z) = x_0 + BG(z)x_Q. \tag{28}$$

For the population observable, $o^T = (o_0^T, 0)$, the standard simple-pole residue [26, 30] becomes

$$a_i = \frac{(o_0^T r_i)[l_i^T b(\lambda_i)]}{l_i^T [I - K'(\lambda_i)] r_i}. \tag{29}$$

At the four crossings $N = 24, 32, 40, 48$, this expression reproduces the archived full-mode residue vectors to relative errors below 2.1×10^{-9}. For the high-branch preparation, retaining the same roots and denominator but replacing b by x_0 gives errors of 3.69, 10.90, 29.91, and 80.46. Correct rates alone do not fix observable weights.

A physical control compares the fold product state ρ_f with its globally excitation-number-dephased state $\rho_d = \sum_k \Pi_k \rho_f \Pi_k$. The two states have identical initial $q = 0$ populations but different coherences. Their difference $\delta\rho_c = \rho_f - \rho_d$ consequently generates a nonzero density transient. A source depending only on x_0 would instead predict zero difference. The two real modes still give a 101% relative error over $6 \le t \le 12$ at the $N = 32$ crossing. Source reconstruction therefore does not justify a two-mode dynamical closure. Section VI compares the complete signal with independent-atom and mean-field controls.

VI. DYNAMICAL REACH AND LIMITS OF THE REAL-MODE PAIR

A. Accuracy of spectral truncations

Large expansion coefficients and preparation-dependent visibility make a direct dynamical comparison necessary [10, 11, 21]. We compare the two-real-mode sum with full evolution for the preparations in Sec. III. At $N = 32$, we also include the oscillatory conjugate pair in Eq. (9). Sparse exponential multiplication supplies the reference without fitting transient amplitudes or rates.

At the crossing, the high-branch relative L^2 error over $6 \le t \le 12$ decreases from 25.1% with two real modes to 0.506% with four modes. The fold-state errors are

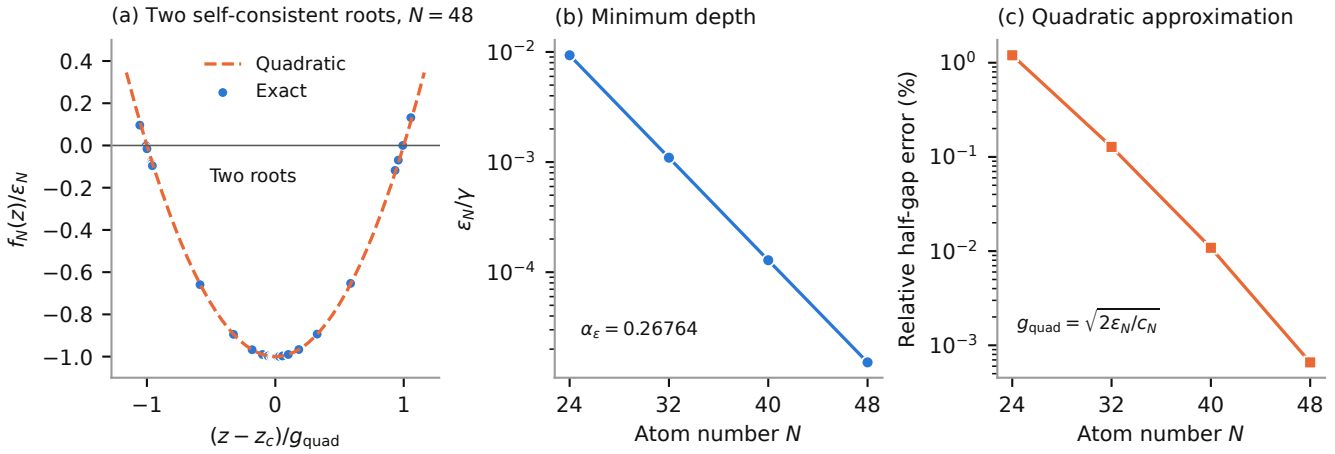


FIG. 4. Exact coherence elimination at the supplied real-axis crossing detunings. (a) The normalized self-consistency function for $N = 48$ (points) and its local quadratic expansion (dashed). (b) Negative-minimum depth and its descriptive four-size exponent. (c) Relative error of the quadratic half-gap. The exact eliminated roots have much smaller errors than this local expansion. Both the depth and curvature are outputs of a finite-N spectral calculation; no independent large-N action is inferred.

69.5% and 5.00%, respectively, demonstrating the preparation dependence of the truncation. The high-branch four-mode error stays below 0.53% at $\delta = \pm 0.06$. The SM displays the spectral contributions after removing the common decay envelope; all errors use the unscaled signals. The oscillatory pair is needed over this time window, while the real-pair mean decay rate remains of order γ.

B. Fixed-preparation tests across the crossing

To distinguish mode exchange from changes in preparation, we freeze the high-branch difference at $\Delta_*(N)$, denoting it by $\delta\rho^*_{\mathrm{hi}}$. We also use the coherence difference $\delta\rho_c$, with the same single-atom fold seed at every size and detuning. For each fixed source x, the complete density signal is

$$s_N(t, \Delta) = \mathrm{Tr}\left[\frac{N_e}{N} e^{tL(\Delta)} x\right]. \tag{30}$$

We compare the two sides $\Delta_* \pm w_N$, where $w_N = 2g_N/|s_a - s_b|$ is the local mixing scale. The scans cover $N = 24, 32, 40, 48$ and sample $t = 0, 0.1, \ldots, 12$, without removing any decay envelope.

For the frozen high-branch source at $N = 48$, individual residues change by approximately 8.5×10^{-4} across $\pm w_N$. The maximum sampled change of the complete density signal is only 1.89×10^{-6} [Fig. 5(c,d)]. Direct propagation of the first two detuning derivatives predicts both endpoint signals from the center, without a fit or spectral truncation. The quadratic prediction has relative L^2 error below 1.1×10^{-5} for this source at all four sizes. The SM gives the corresponding coherence-source tests and the elementary two-pole and trace-contraction bounds. These comparisons support smooth local response despite strong redistribution of spectral weights.

The coherence-source peak also requires a baseline. At $\Delta_*(32)$, its largest sampled magnitude is 0.1343, compared with 0.1636 for independent atoms at the same bare detuning [Fig. 5(a)]. Holding that detuning and the single-atom seed fixed gives peaks between 0.134 and 0.135 for $N = 8$–48 [Fig. 5(b)]. Thus this peak is neither enhanced over independent atoms nor evidence of increasing response with size in this control. The global initial trace distance is not held fixed across sizes (SM).

Mean field nevertheless fails to reproduce the complete coherence-source transient [Fig. 5(a)]. Its control evolves each common-phase product state separately before averaging, retaining the finite-N interaction factor $(N-1)/N$. The discrepancy cannot be assigned to the target pair, whose standalone error for this preparation is already large. The negative enhancement and narrow-response tests apply to the stated preparations, density observable, sizes, and time window; they do not exclude other protocols.

VII. DISCUSSION AND CONCLUSIONS

The locally damped Rydberg ensemble exhibits a finite-size exchange of relaxation residues without a corresponding strong, narrow density response in the tested fixed preparations. This connects mode character, exact coherence elimination, and full transients within one model. Earlier collective-spin studies established finite-rate doublets and cancelling residues [20, 21]; the present calculation resolves their detuning-dependent response counterpart with all local-decay spin blocks retained.

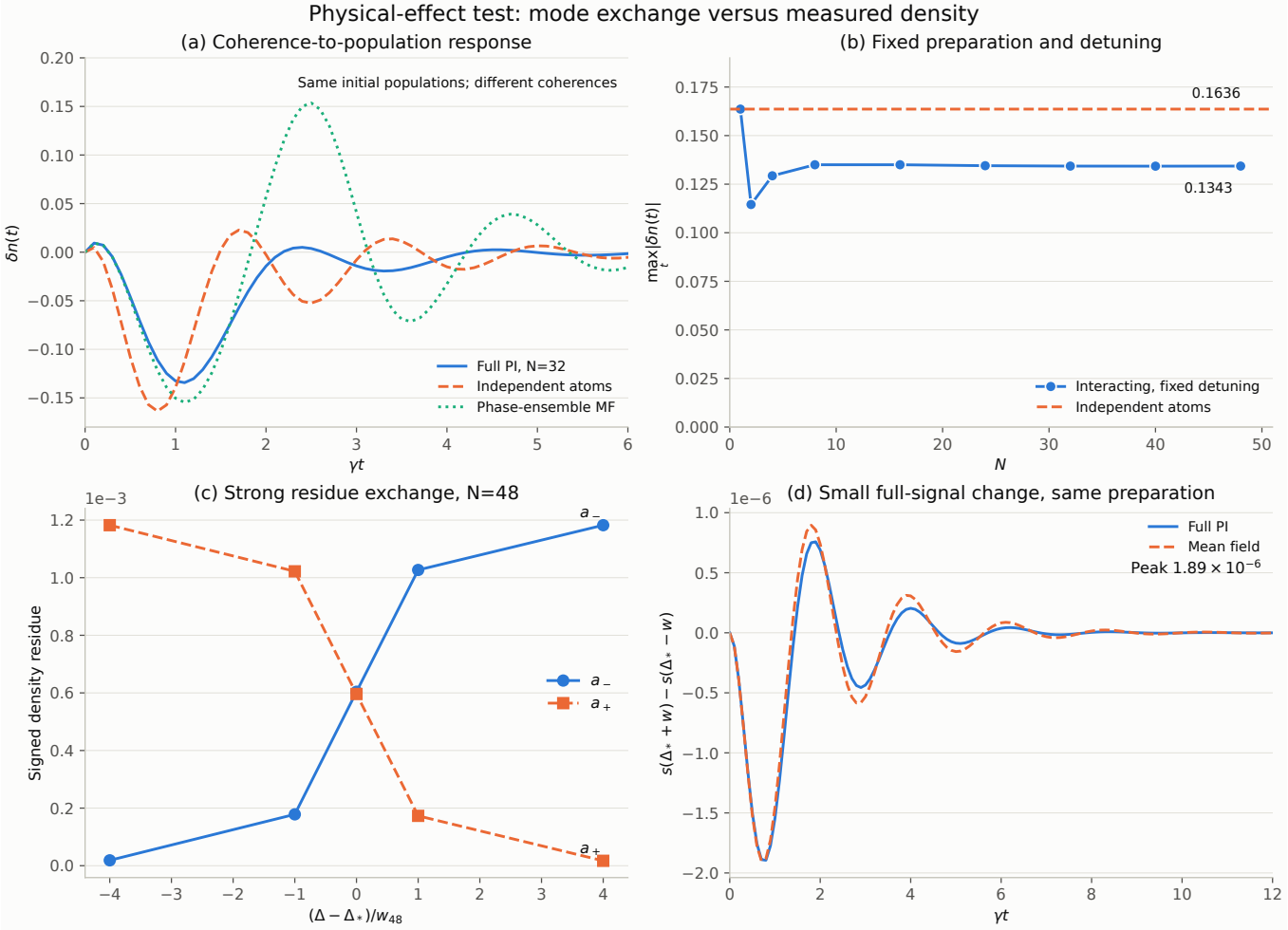


FIG. 5. Fixed-preparation density controls, in raw density units. (a) Fold coherence difference at $N = 32$, $\Delta = \Delta_*(32)$: full permutation-invariant evolution, independent atoms, and a common-phase mean-field ensemble. (b) Largest sampled magnitude at the same detuning and single-atom seed, for $t = 0, 0.1, \ldots, 12$. (c) Signed real-mode residues for the $N = 48$ high-branch difference frozen at its crossing. (d) Difference between the complete signals at $\Delta_* + w_{48}$ and $\Delta_* - w_{48}$ for that fixed preparation. Lines in (b,c) guide the eye; residues in (c) are decomposition weights, whereas (d) compares physical density signals.

The initial-coherence source is essential to that connection, because correct decay roots alone do not determine observable weights.

The independent-atom and fixed-detuning controls restrict the physical interpretation of the coherence-to-population peak. Its amplitude does not identify the near-degenerate pair, even though the complete transient differs substantially from mean field. Thus the spectral exchange is a statement about the decomposition of relaxation, not evidence for enhanced population switching. Identifying the many-body origin of the remaining mean-field discrepancy requires resolving spectral contributions beyond this pair. Other preparations, observables, or time-dependent protocols remain outside the present tests.

An asymptotic explanation would require the scaling of ϵ_N/c_N, or equivalently $V_{ab}V_{ba}$, independently of the exact pair. Exact steady-state information is available [12], but the barrier conjecture of Noachtar and Clerk [16] addresses the ordinary Liouvillian gap through a specified purification. It does not predict this finite-rate splitting from an arbitrary excitation-number marginal potential. Applying switching-instanton theory [14, 15] likewise requires boundary conditions appropriate to this excited spectral pair. The reported finite-window exponents establish neither derivation nor a thermodynamic fold correspondence.

DATA AND NUMERICAL METHODS

The SM specifies the generator, preparations, projection identities, control calculations, and numerical checks. Plotted records, failed numerical attempts and their refinements, and reproduction scripts accompany

the source bundle. Independent full-spin checks cover $N = 1$–4, and detuning derivatives are checked against independent small-system time integration. The finite-size calculations use no fitted stochastic noise model.

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